TÉCNICO LISBOA

Robótica de Manipulação
T1 - Introduction

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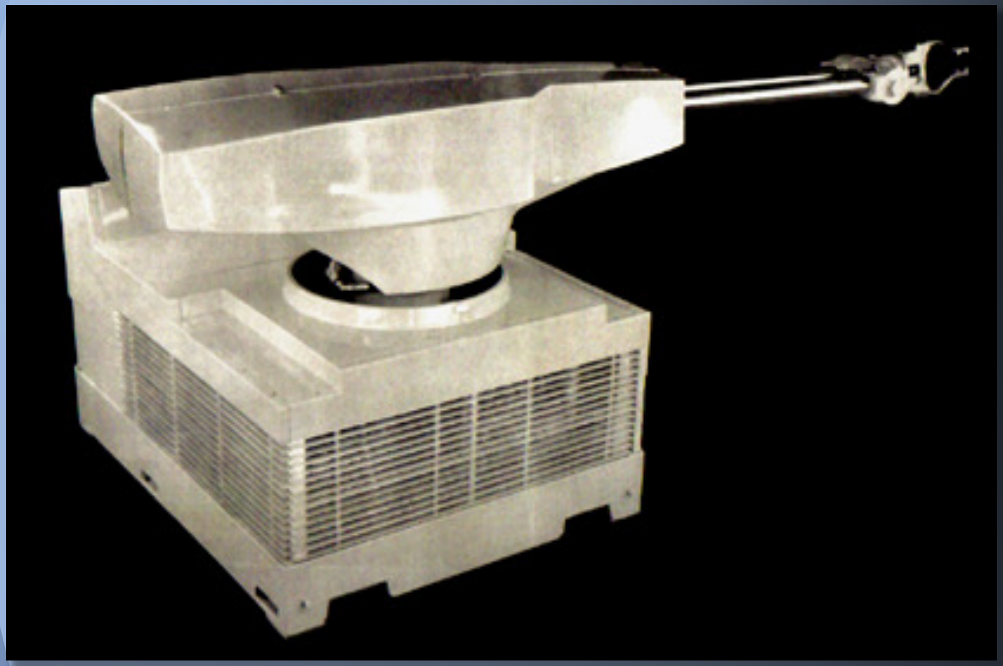
Professor Associado



The first industrial robot
The UNIMATE
Credit Mr. George C. Devol, Inventor

Departamento de Engenharia Mecânica
Área Científica de Controlo, Automação e Informática Industrial

DEM
Controlo, Automação e Informática Industrial

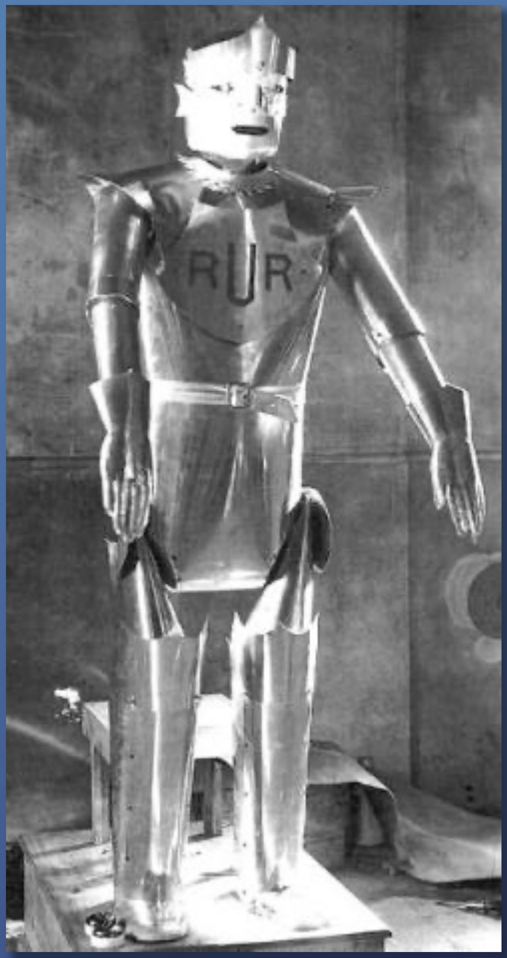
Robotics

Human beings have constantly attempted to seek substitutes that would mimic their behavior in the various instances of interaction with the surrounding environment.

It is in fiction that the most spectacular substitutes have been idealized, and those have heavily motivated the current state of the art in Robotics.

Evidences of this quest are found in:

-Greek Mythology
-Mary Shelly's Frankenstein
-Karel Kapek's Rossum's Universal Robots, 1920
First use of the term robot
-Isaac Asimov's science fiction stories, 1940s
The term robotics was introduced



Robotics

In Isaac Asimov's science fiction, the term *robotics* was introduced as the science devoted to the study of robots which was based on the three fundamental laws:

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey orders given to it by human beings, except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

Isaac Asimov

The Caves of Steel, p.177-179, 1942

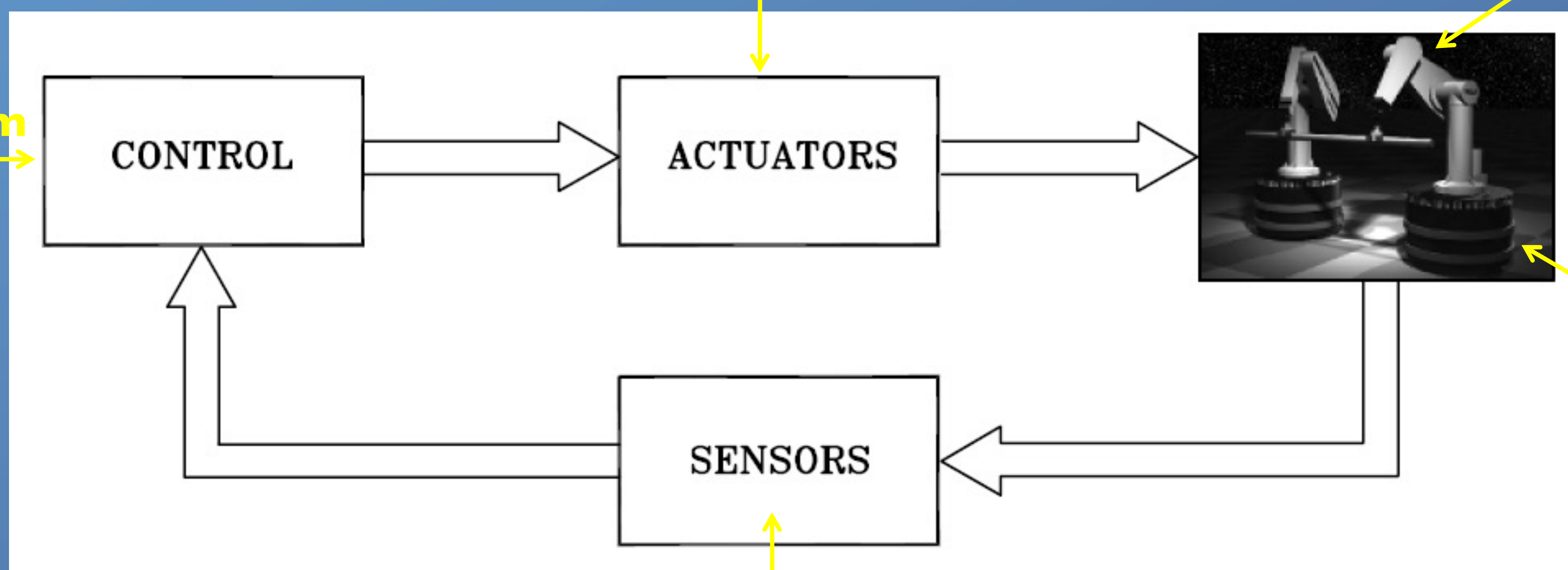
Robotics

In science, *robotics* is commonly defined as the science studying the intelligent connection between perception and action.

Actuation System
(servomotors, drives and transmissions)

Manipulation apparatus

Control System



Locomotion apparatus

Sensory System
(proprioceptive and exteroceptive sensors)

Robotics: interdisciplinary field of mechanics, control, computers and electronics

Robot Mechanical Structure

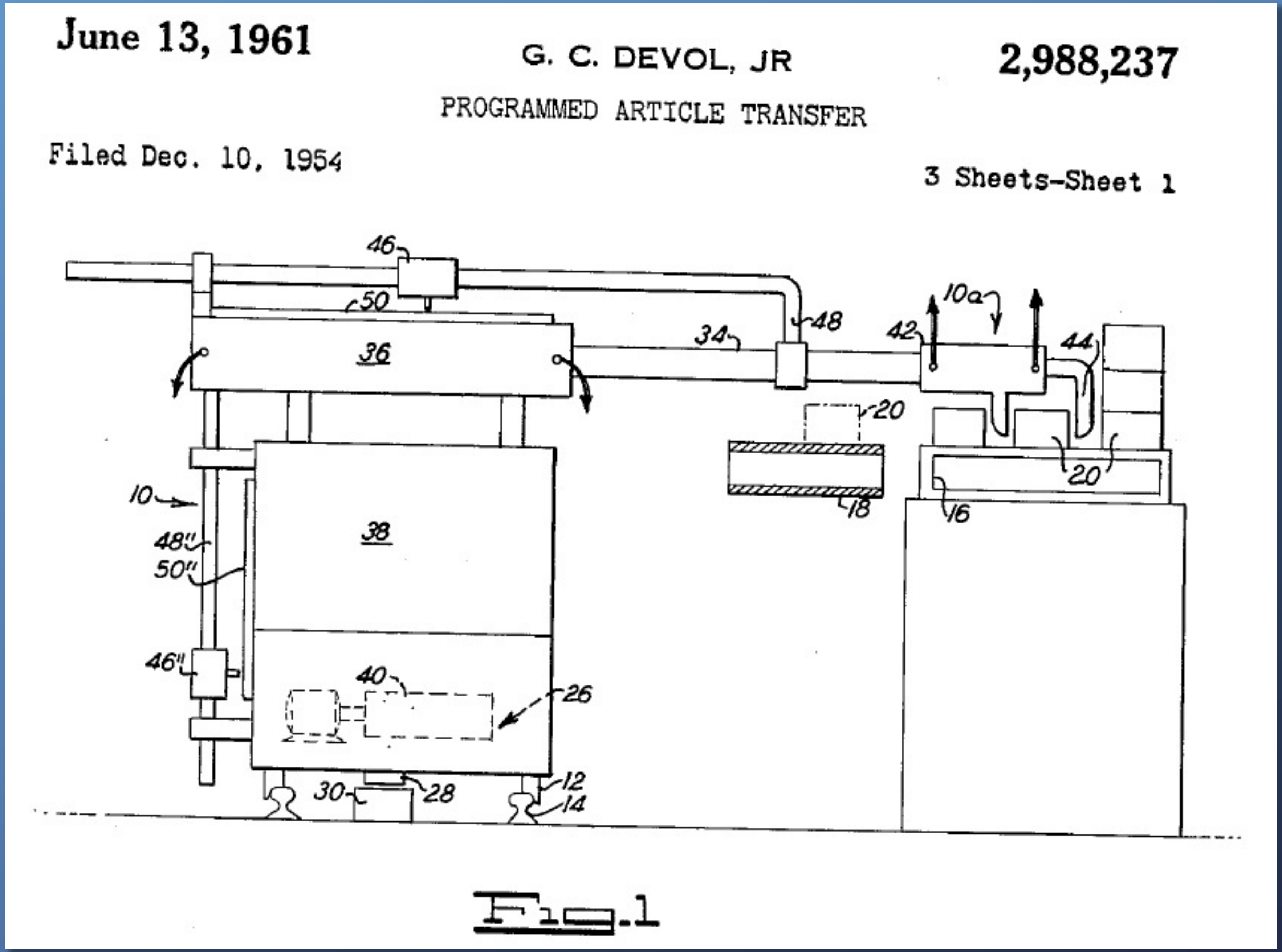
The key feature of a robot is its mechanical structure:

Robot Manipulators: robots with a fixed base,

Mobile Robots: robots with a mobile base.

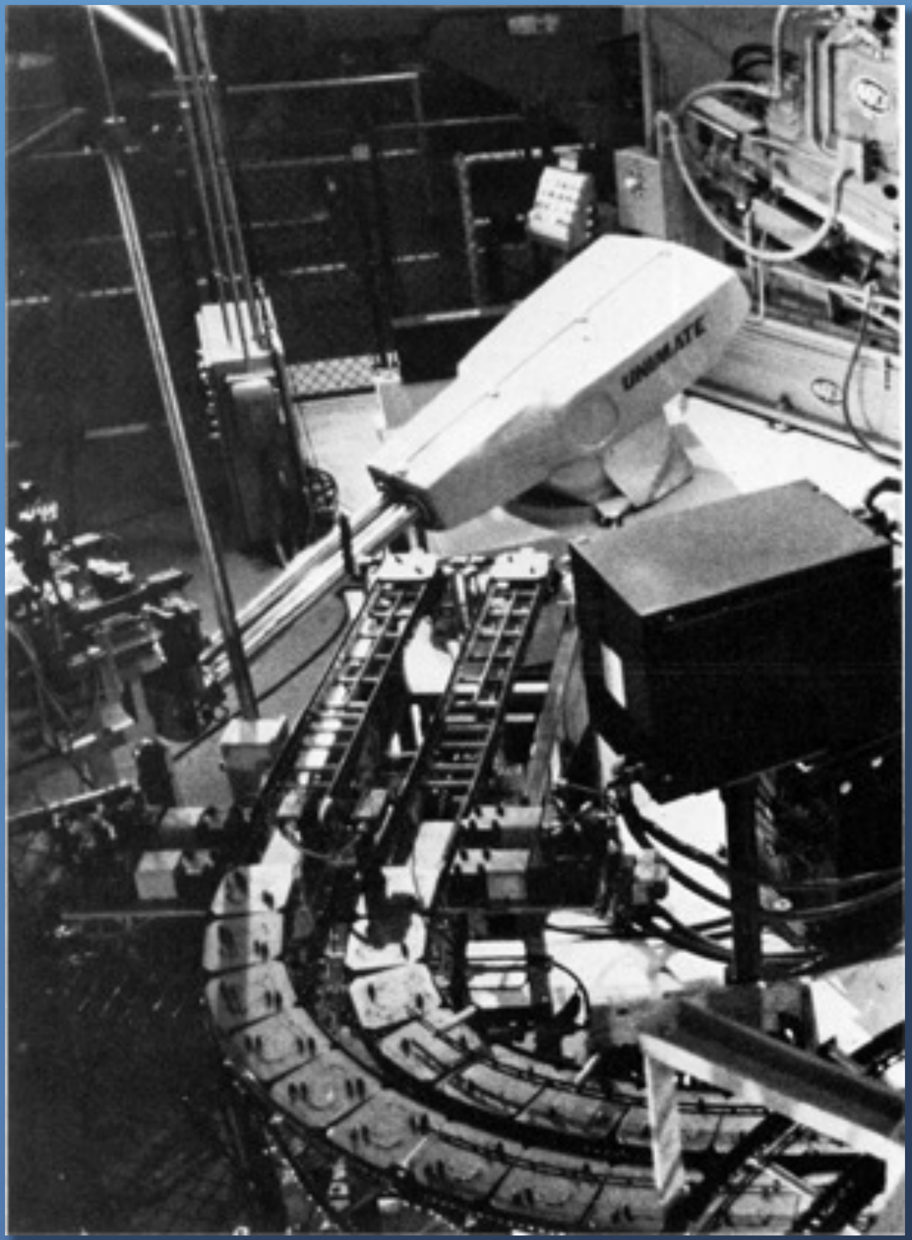
The first industrial robot manipulator – UNIMATE

In 1954, **George C. Devol** applied for US Patent No. 2,988,237 for *Programmed Article Transfer*, which was issued in 1961.



The first industrial robot – UNIMATE

Based on his patent, George Devol joined Joseph Engelberger to form **Unimation Incorporated**. In 1961, it's first product, the UNIMATE, was sold to GM, where it was used for die casting handling.



Obeying step-by-step commands stored on a magnetic drum, the hydraulic 1.8 ton arm is versatile enough to perform a variety of tasks, and manipulates loads up to 200 Kg.

Approximately five million dollars were spent to develop the first UNIMATE. The company only showed its first profit in 1975.

2003 Inductee of the Robot Hall of Fame
(Carnegie Mellon University)
http://www.robothalloffame.org/

Industrial robot as defined by ISO 8373:
Source: International Federation of Robotics (www.ifr.org)

An automatically controlled, reprogrammable, multipurpose manipulator programmable in three or more axes, which may be either fixed in place or mobile for use in industrial automation applications.

Reprogrammable: whose programmed motions or auxiliary functions may be changed without physical alterations;

Multipurpose: capable of being adapted to a different application with physical alterations;

Physical alterations: alteration of the mechanical structure or control system except for changes of programming cassettes, ROMs, etc.

Axis: direction used to specify the robot motion in a linear or rotary mode

Industrial robots

The AdeptOne XL robot

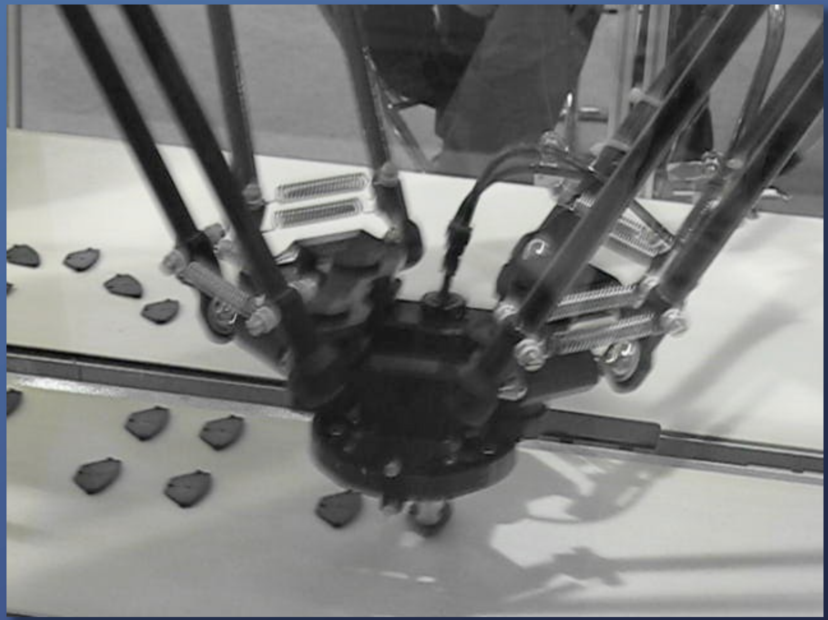
Repeatability: 0.025mm
Payload: 12 Kg

The ABB IRB140

Repeatability: 0.03mm
Payload: 6 Kg

The ABB IRB360

Repeatability: 0.1mm
Payload: 1 Kg

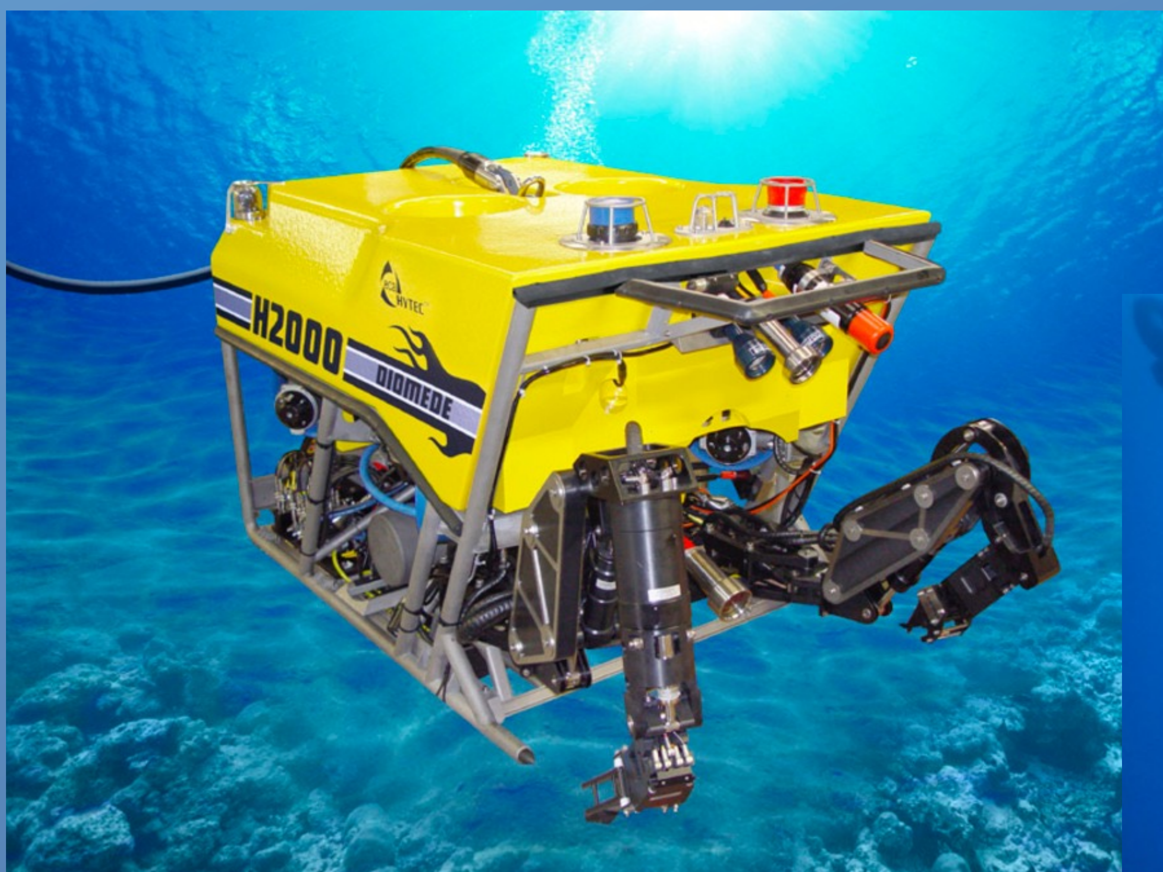


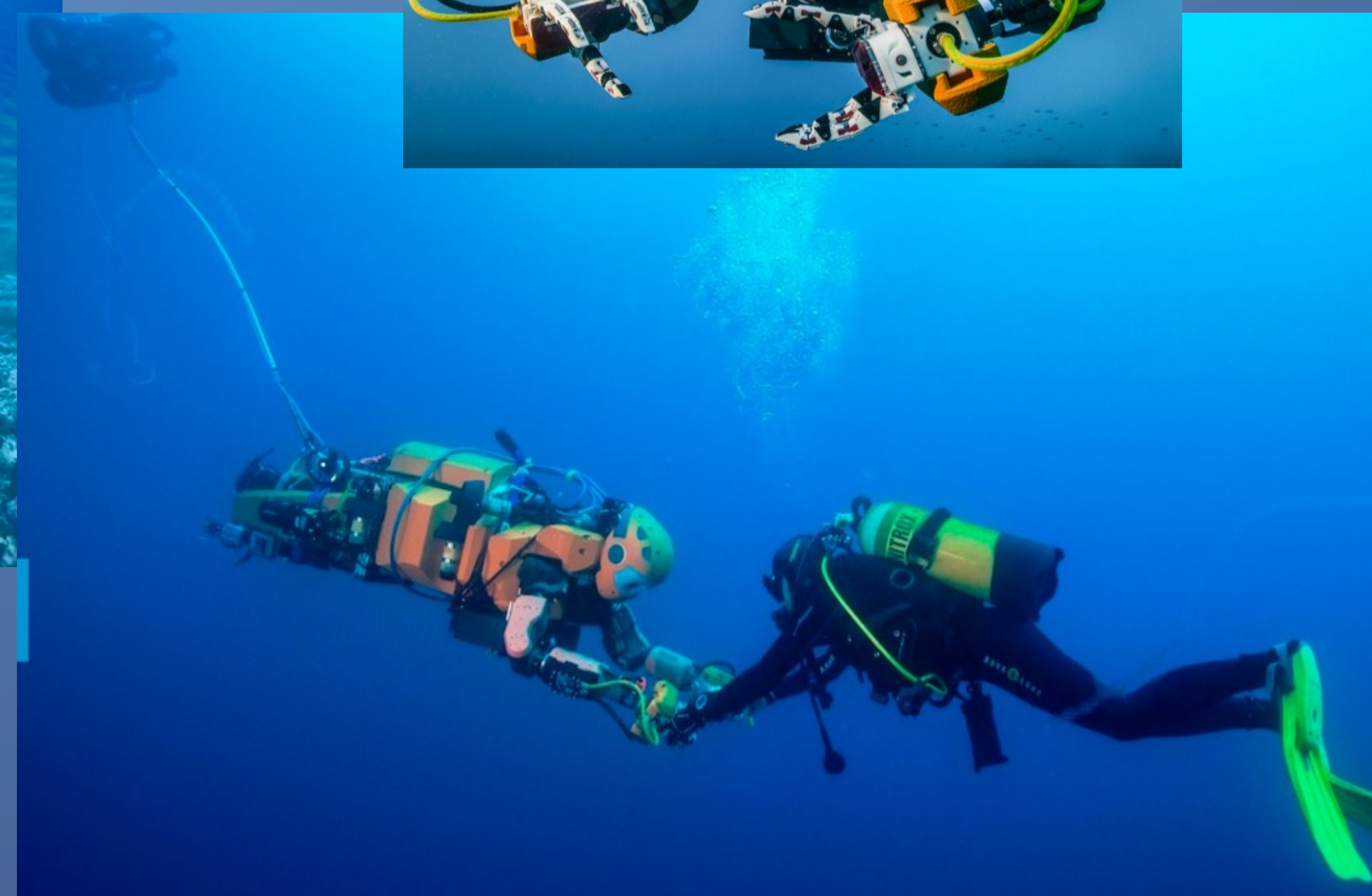
Robots in the Energy Industry

"The growing demand in alternative energy sources after the recent nuclear catastrophe in Japan will push robot installations, e.g. the production of solar cells etc."

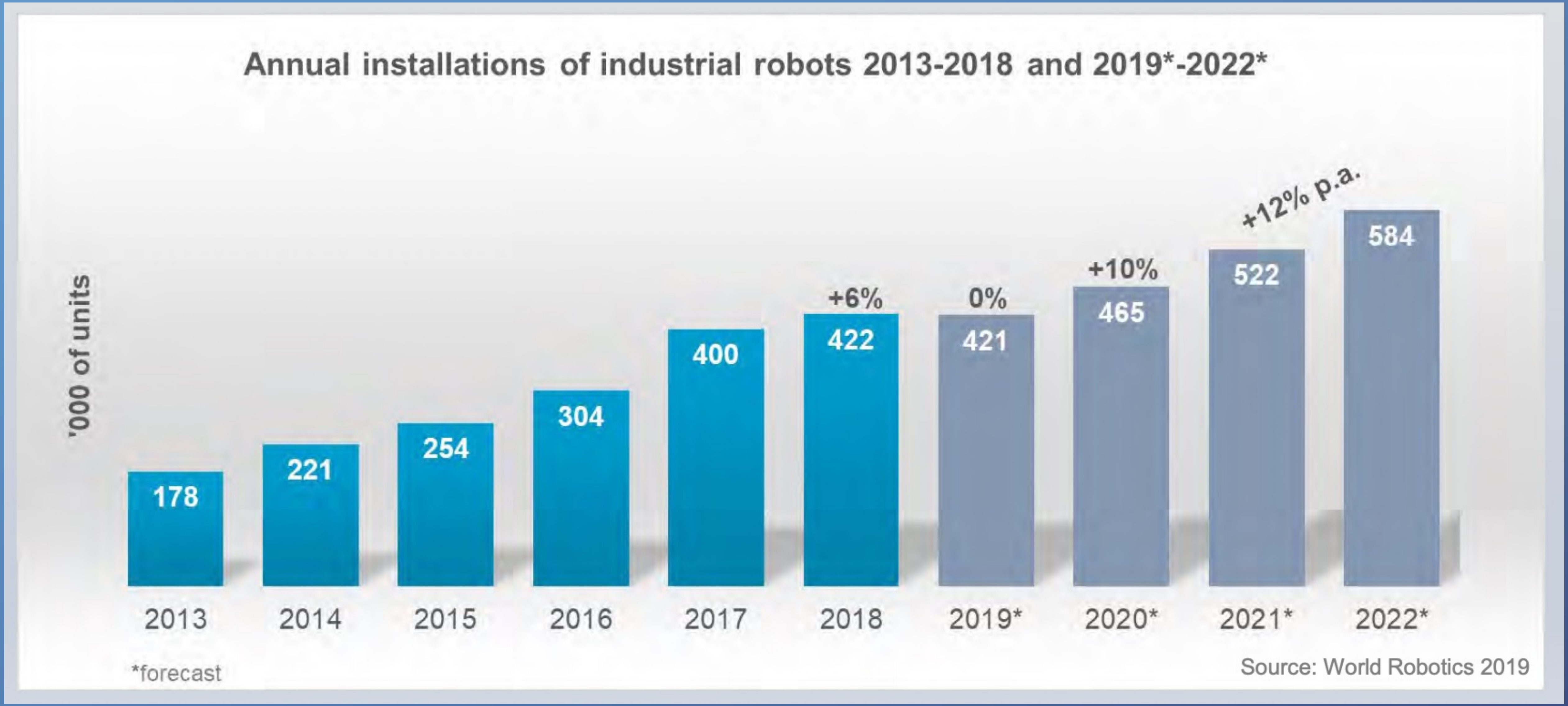
- International Federation of Robotics (www.ifr.org)

https://spectrum.ieee.org/automaton/robotics/humanoids/stanford-humanoid-submarine-robot

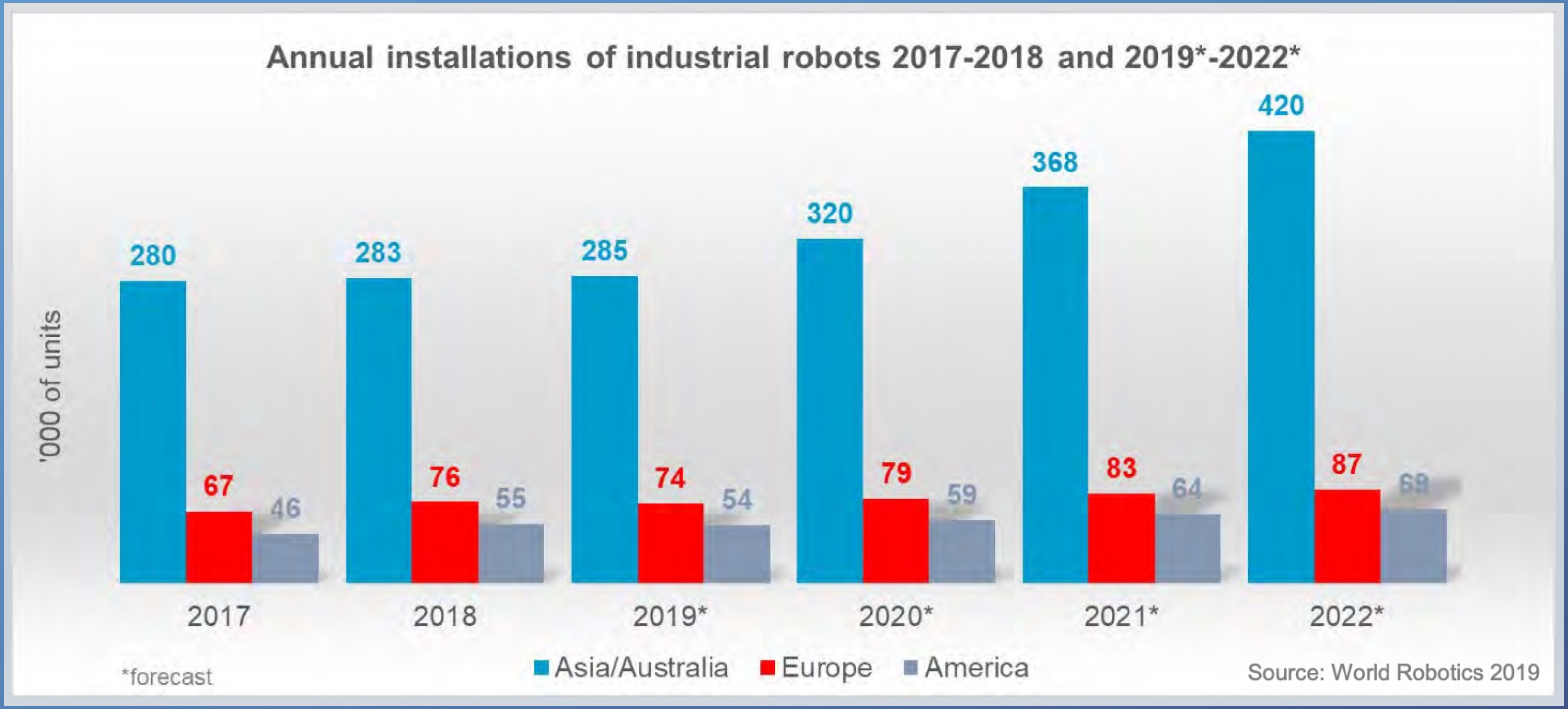




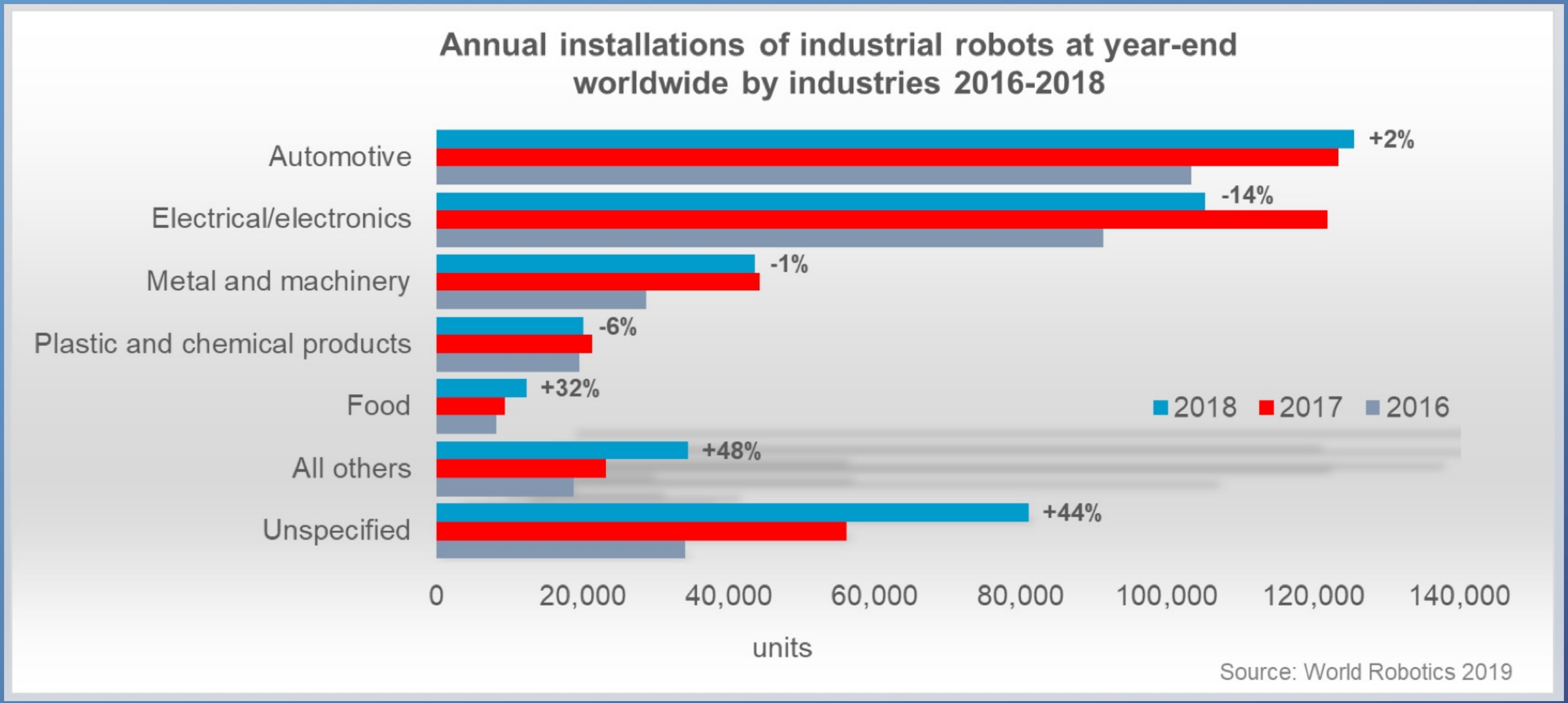
Industrial robots
Source: International Federation of Robotics (www.ifr.org)



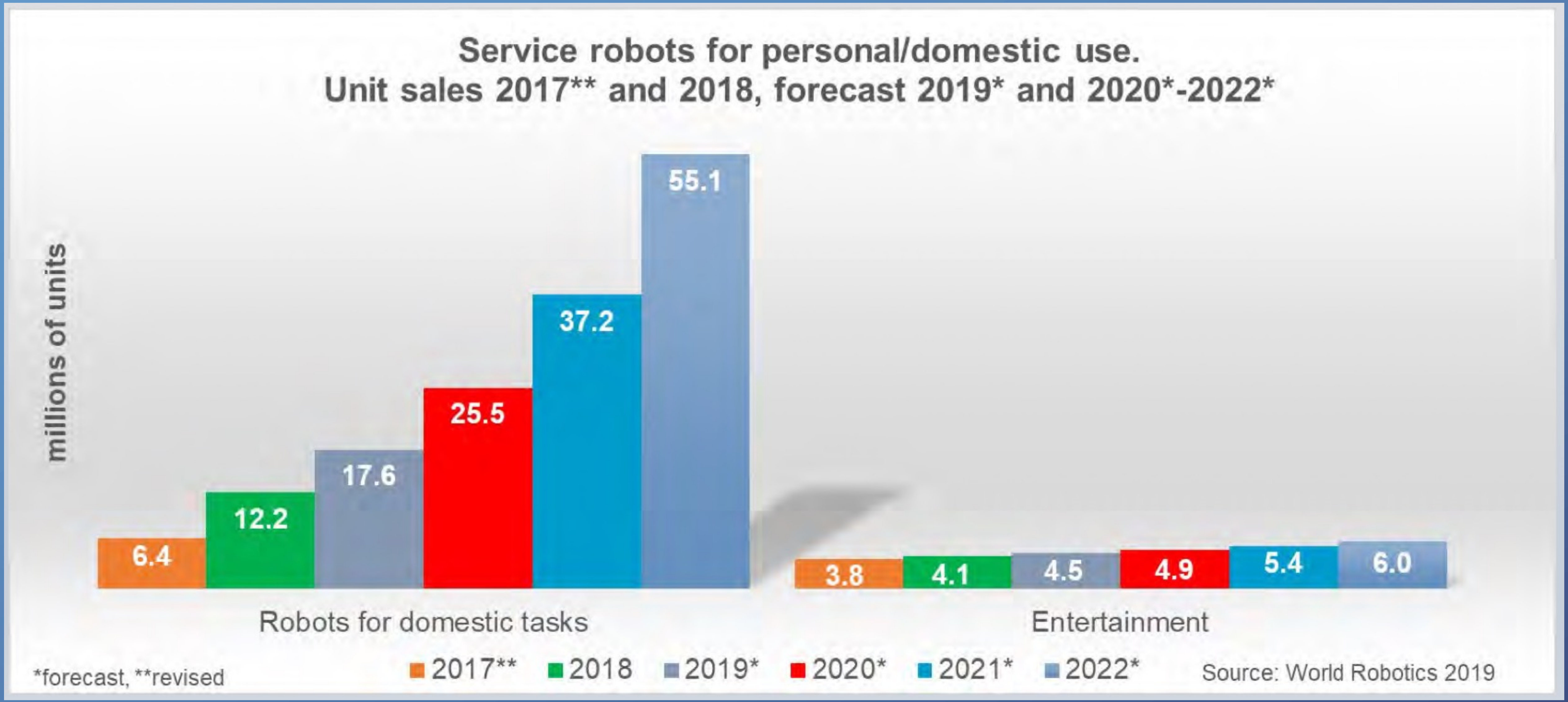
Industrial robots
Source: International Federation of Robotics (www.ifr.org)



Industrial robots
Source: International Federation of Robotics (www.ifr.org)



Service Robots
Source: International Federation of Robotics (www.ifr.org)



Service robots (www.ifr.org)

Definition of Service Robots

In a joint effort started in 1995 the United Nations Economic Commission for Europe (UNECE) and IFR engaged in working out a preliminary service robot definition and classification scheme, which has been absorbed by the current ISO Technical Committee 184/Subcommittee 2 resulting in a novel ISO-Standard 8373 which had become effective in 2012. A preliminary extract of the relevant definitions is given here:

A robot is an actuated mechanism programmable in two or more axes with a degree of autonomy, moving within its environment, to perform intended tasks. Autonomy in this context means the ability to perform intended tasks based on current state and sensing, without human intervention.

A **service robot** is a robot that performs useful tasks for humans or equipment excluding industrial automation application. Note: The classification of a robot into industrial robot or service robot is done according to its intended application.

A **personal service robot or a service robot for personal use** is a service robot used for a non-commercial task, usually by lay persons. Examples are domestic servant robot, automated wheelchair, personal mobility assist robot, and pet exercising robot.

A **professional service robot or a service robot for professional use** is a service robot used for a commercial task, usually operated by a properly trained operator. Examples are cleaning robot for public places, delivery robot in offices or hospitals, fire-fighting robot, rehabilitation robot and surgery robot in hospitals. In this context an operator is a person designated to start, monitor and stop the intended operation of a robot or a robot system.

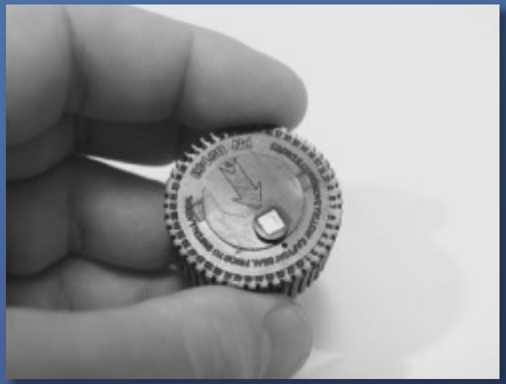
A robot system is a system comprising robot(s), end-effector(s) and any machinery, equipment, devices, or sensors supporting the robot performing its task.

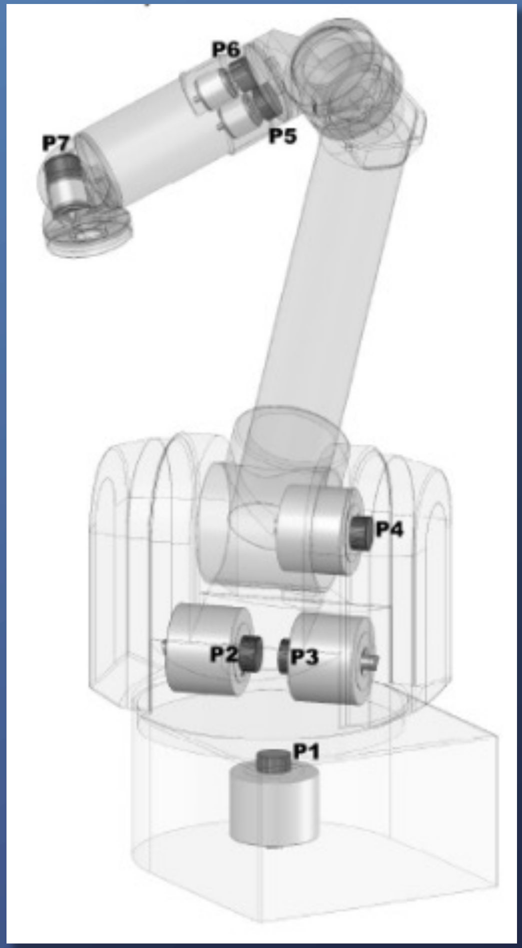
Lightweight robots

WAM – Whole Arm Manipulator
Barrett Technology
(www.barretttechnology.com/robot/)

Manipulator weight: 27Kg
Payload: 3Kg
Endtip Velocity: 1 m/s

Joints: Cable-differential, small sized brushless motors – PUCK servodrives





Rooks, B., "The harmonious robot", *Industrial Robot: An International Journal*, 33(2):125-130.

Lightweight robots

Medical Robotics – DLR MIRO
Universal surgical robot

MI Surgery with force feedback

Lightweight robots

Manipulator weight: 14Kg
Payload: 14Kg
Max joint speed: 120°/s

Link material: Carbon fiber composite (exoskeleton structure)

Joints: Brushless dc motors and Harmonic Drive reduction gears, with integrated electronics

Elasticity and high friction at the joints due to the *flexspline* of the reduction gears

Lightweight robots

DLR – LWR JUSTIN

Robot Manipulators

The mechanical structure of a manipulator consists of a sequence of rigid bodies (links) interconnected by means of articulations (joints), forming a serial or kinematic chain.

From a topological viewpoint, the kinematic chain is termed open when there is only one sequence of links connecting the two ends of the chain. When a sequence of links forms a loop, the kinematic chain is termed closed.

The joints may be either revolute, or prismatic.

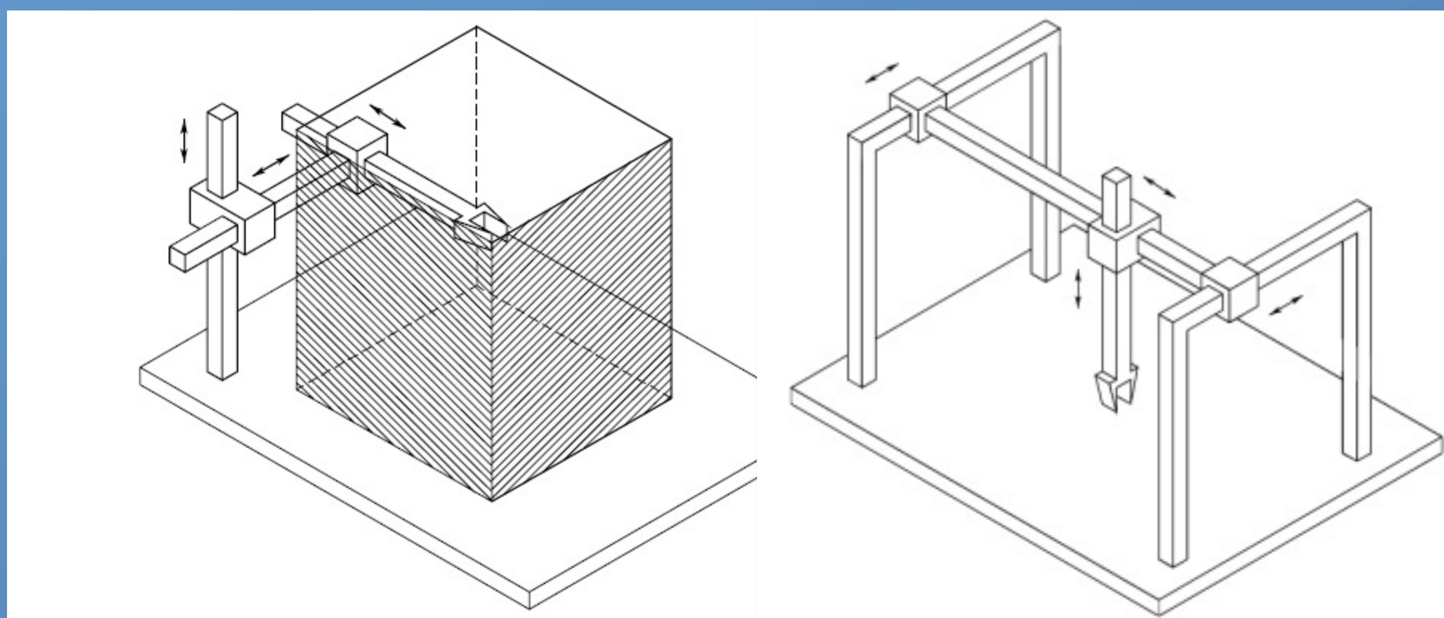
A manipulator is characterized by an arm that ensures mobility, a wrist that confers dexterity and an end-effector that performs the task required of the robot.

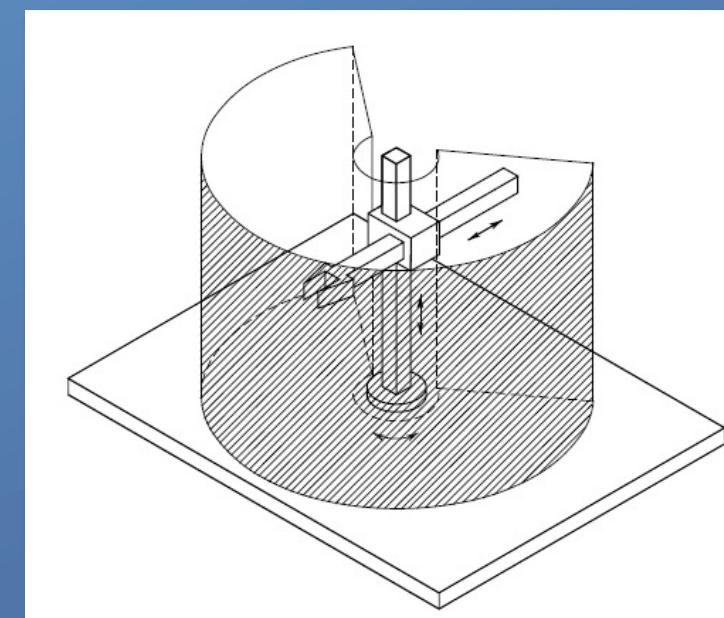
For arbitrary positioning and orienting an object in 3D space, six degrees of freedom are necessary.

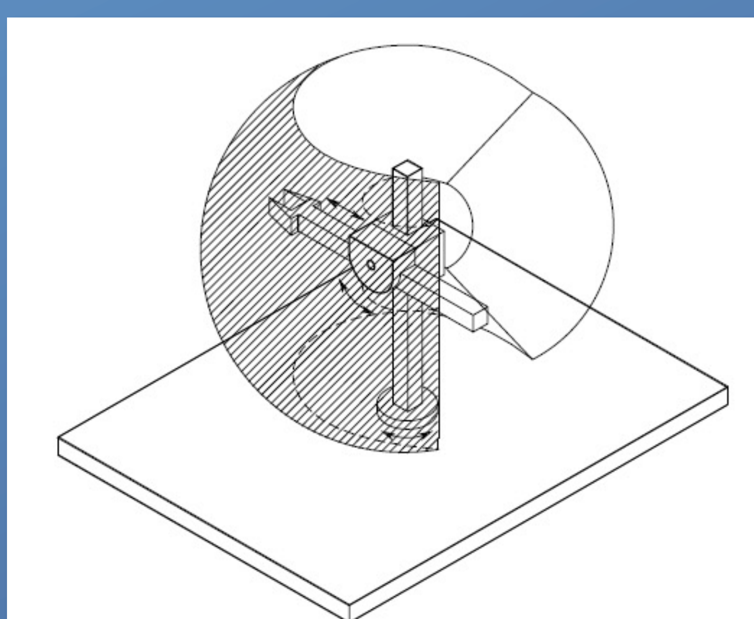
The workspace represents that portion of the environment that the manipulator's end-effector can access.

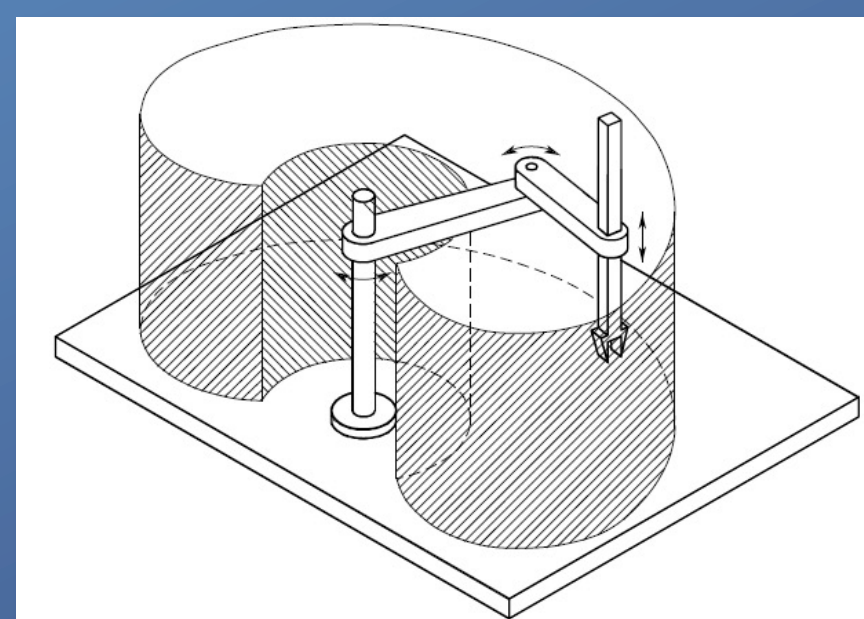
Manipulator Structures

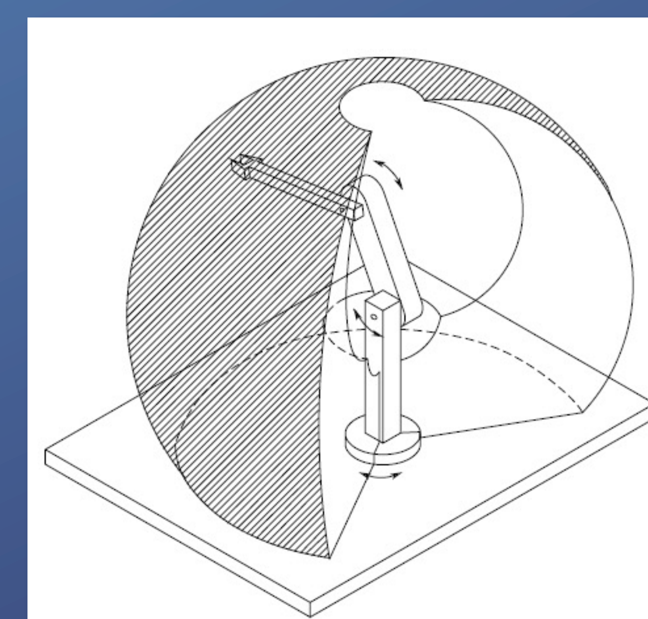
The type and sequence of the arm's joints, starting from the base joint, allows a classification of manipulators as **Cartesian**, **Cylindrical**, **Spherical**, **SCARA** and **anthropomorphic**:





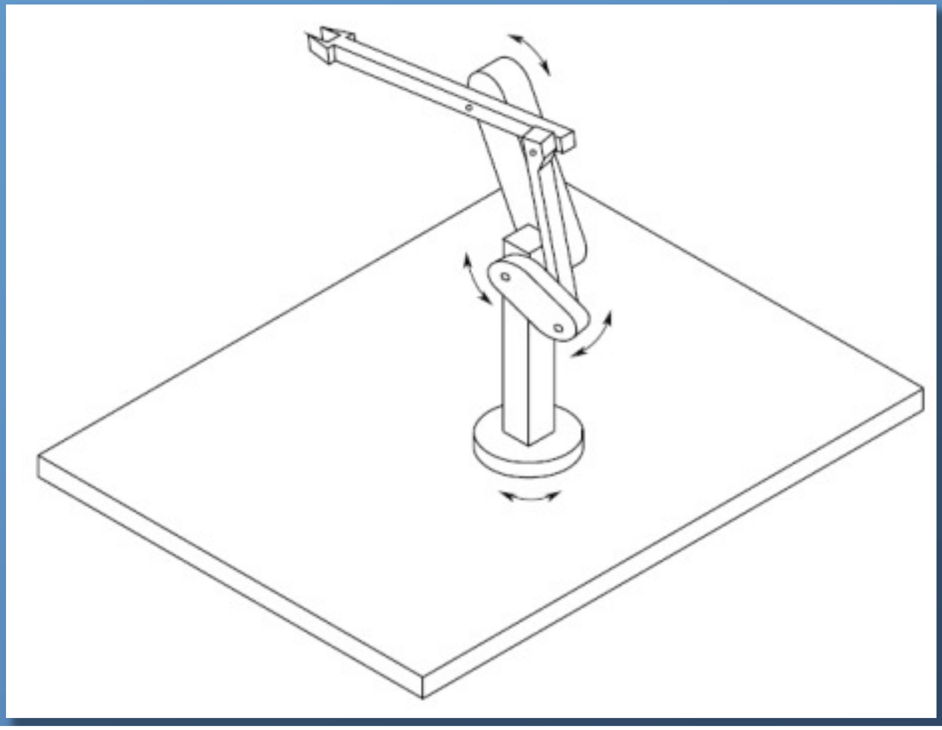


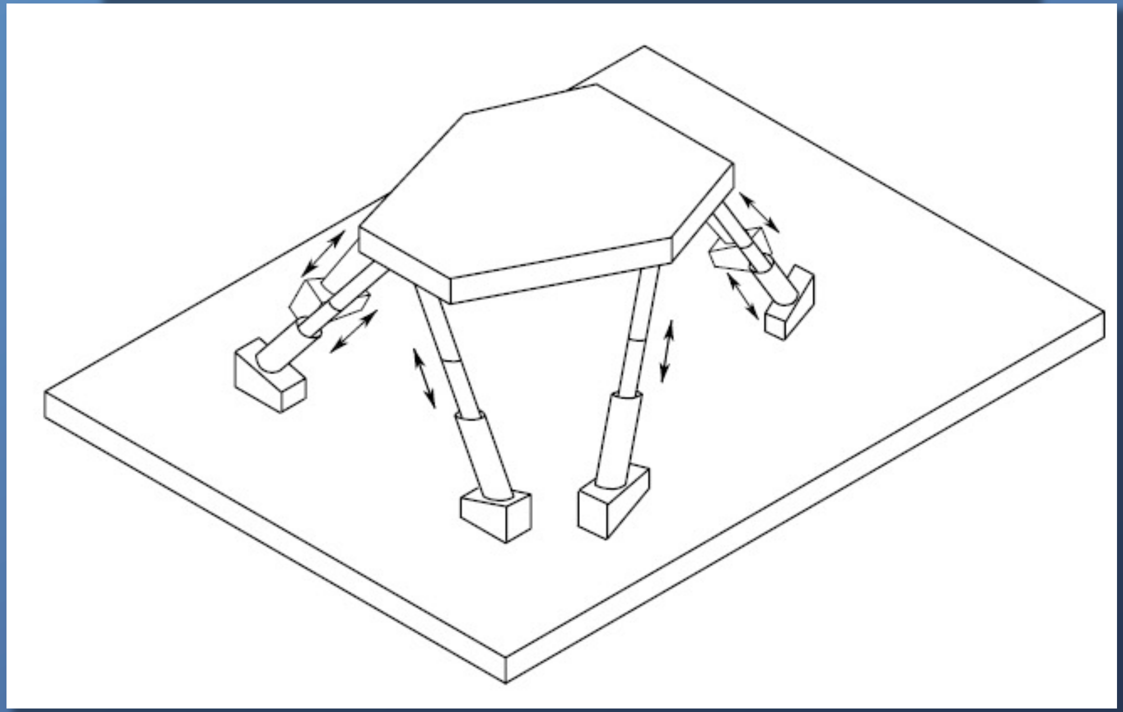


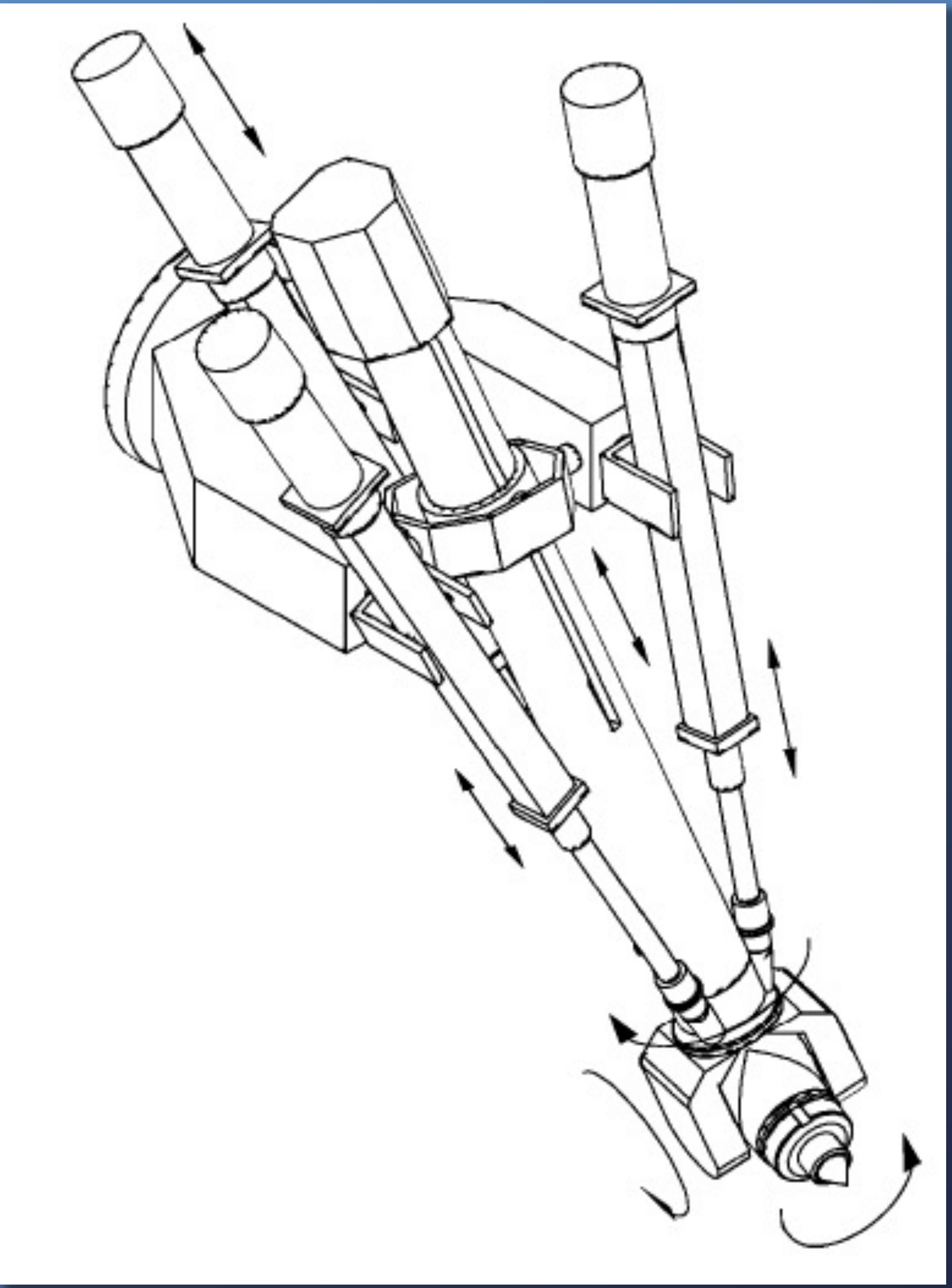


Manipulator Structures

When higher payloads are required, the mechanical structure will have higher stiffness in order to guarantee comparable accuracy.



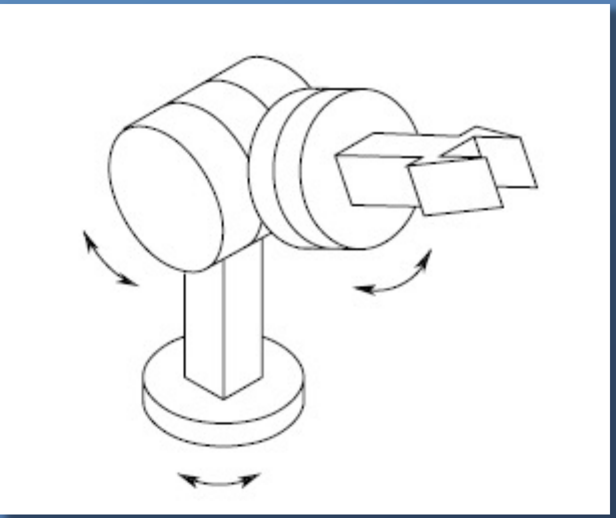




Wrist

The arm structures are required to position the wrist which is then required to orient the manipulator's end effector.

The structure that renders the highest dexterity and compactness is one with three revolute axis intersecting at a single point: the spherical wrist



The key feature of a spherical wrist is the decoupling between position and orientation of the end effector – the arm positions the wrist point of intersection, and the wrist determines the end effector orientation.